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OF APPLIED SCIENCES

Electronic Circuit Design

Lecture notes

Chapter 5: Bridge Circuits

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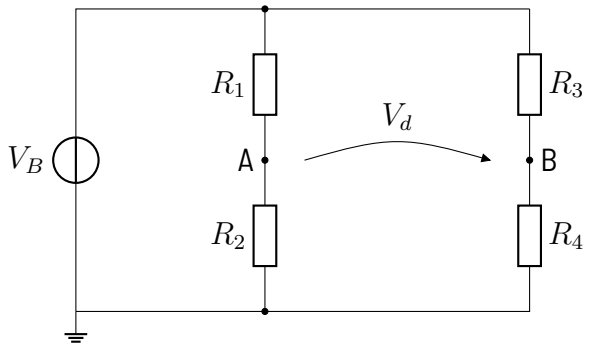
5 Bridge Circuits

5.1 The Wheatstone Bridge

Many sensors are resistors whose value changes a little with the measured quantity: a platinum resistor changes by $0.385\,\%$ per kelvin, a strain gauge by $0.2\,\%$ at the elastic limit of steel. If we put such a sensor into a simple voltage divider, the interesting change is a tiny ripple on a large constant voltage. Amplifying it would amplify the constant part as well, and any drift of the supply would look like a change of the measured quantity.

The bridge circuit solves this problem with a second voltage divider that produces the constant part on its own. The difference between the two divider outputs is zero when the sensor has its reference value and contains only the change. The circuit was described by Christie in 1833 and made popular by Wheatstone in 1843 as a method to measure an unknown resistance by adjusting a known one until the galvanometer between the two dividers shows zero (*null method*). Today the bridge is mostly used in the *deflection mode*: the resistors are fixed and the bridge output voltage is amplified and digitised. This is the first stage of the capstone task [2, 6].

Two voltage dividers



Bridge output

$$V_d = V_A - V_B = V_B \left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right)$$

Here V_B is the excitation (supply) voltage of the bridge. Each leg is an unloaded voltage divider from chapter 1; the output is the difference of the two tap voltages. The output is taken between two nodes of which neither is at ground: a *differential* voltage.

Balance condition

- the bridge is **balanced** if $V_d = 0$:

Balance condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad \Leftrightarrow \quad R_1 R_4 = R_2 R_3$$

- the balance does **not** depend on V_B: supply noise and drift do not appear at the output
- null method: adjust a known resistor until $V_d = 0$, then $R_2 = R_1 R_4 / R_3$
- deflection method: fixed resistors, balanced at the reference point (e.g. $0\,^\circ\text{C}$), measure V_d

At balance the output is zero, independent of the supply. Away from balance the output is proportional to V_B: the bridge converts a relative resistance change into a fraction of the excitation voltage. This is why the excitation must be a stable reference, or the measurement must be ratiometric (chapter 4).

5.2 Output Voltage and Linearity

In most applications only one resistor of the bridge is a sensor (*quarter bridge*). We write its value as $R_2 = R_0(1 + x)$, where $x = \Delta R / R_0$ is the relative change, and choose the other resistors such that the bridge is balanced at $x = 0$. The output voltage is then a function of x only. For small x it is proportional to x, for larger x it bends: the bridge is not linear. The reason is that the current in the sensor leg changes with the sensor resistance.

Quarter bridge with equal resistors

- $R_1 = R_3 = R_4 = R_0$, sensor $R_2 = R_0(1+x)$:

$$V_d = V_B\left(\frac{1+x}{2+x} - \frac{1}{2}\right) = \frac{V_B}{4}\cdot\frac{x}{1+x/2}$$

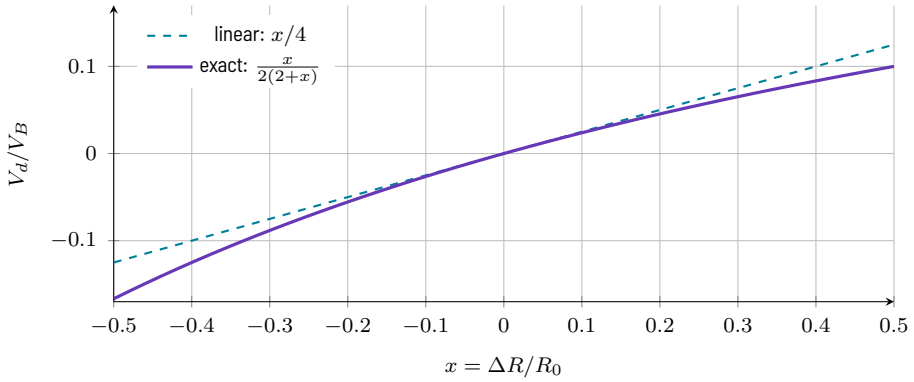
Quarter bridge, small signal

$V_d \approx \dfrac{V_B}{4}\dfrac{\Delta R}{R_0}$ relative error $\approx -\frac{x}{2}$

- example: $V_B = 5\,\text{V}$, $x = 1\,\%$: $V_d = 12.44\,\text{mV}$ (estimate $12.5\,\text{mV}$)
- $x = 50\,\%$: $V_d = 0.500\,\text{V}$ instead of $0.625\,\text{V}$

▶ Falstad: `examples/bridges/wheatstone.txt`

The bridge is not linear



- positive x: output too small, negative x: too large
- strain gauges ($x < 0.5\,\%$): negligible; Pt100 over $100\,\text{K}$ ($x = 38.5\,\%$): $16\,\%$

Unequal arms: trade sensitivity for linearity

- $R_1 = R_3 = m\,R_0$, $R_4 = R_0$, sensor $R_0(1+x)$:

$$V_d = V_B\,\frac{m\,x}{(m+1)(m+1+x)} \approx V_B\,\frac{m}{(m+1)^2}\,x$$

- sensitivity $m/(m+1)^2$: maximum $1/4$ at $m = 1$; $m = 39$: 0.024
- relative nonlinearity $\approx -x/(m+1)$: $m = 39$ makes it 20 times smaller
- large m also limits the sensor current: $I = V_B/(R_1 + R_{\text{sensor}})$ (self-heating!)

A large ratio m turns the sensor leg into an almost ideal current source: the current hardly changes with the sensor resistance, so the voltage across the sensor is nearly proportional to its resistance. The price is a smaller output. Since the amplifier needs a gain anyway, a few tens of millivolts are easy to handle, while a nonlinearity would have to be corrected in software. Other ways to linearise a bridge are a constant current excitation or an op-amp that keeps the bridge at balance [6].

Exercise 5.1: Wheatstone bridge

A bridge with $R_1 = R_3 = R_4 = 1\,\text{k}\Omega$ is supplied with $V_B = 5\,\text{V}$. The resistor $R_2 = R_x$ is the sensor.

a) Calculate V_d exactly and with the small-signal formula for $R_x = 1010\,\Omega$.

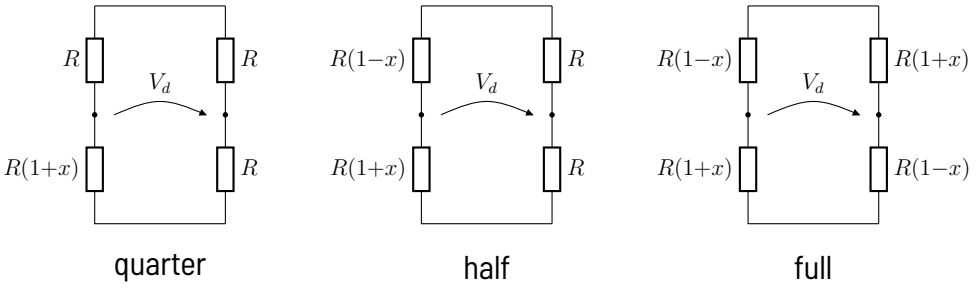
b) Repeat for $R_x = 1500\,\Omega$. How large is the error of the formula?

c) Check both values in Falstad.

5.3 Half and Full Bridges: Strain Gauges

A strain gauge is a thin metal foil meander on a plastic carrier, glued onto a mechanical part. When the part is stretched, the conductor becomes longer and thinner and its resistance rises. The relative change is proportional to the strain $\varepsilon = \Delta l / l$; the factor is the *gauge factor* k, about 2 for metal foil gauges. Typical values are $120\,\Omega$, $350\,\Omega$ and $1\,\mathrm{k\Omega}$; a strain of $1000\,\mu\mathrm{m/m}$ ($0.1\,\%$, close to the elastic limit of steel) changes the resistance by only $0.2\,\%$. On a bending beam, gauges on the top are stretched and gauges on the bottom are compressed by the same amount. Placing them in adjacent arms of the bridge doubles or quadruples the signal and makes the bridge exactly linear. Load cells of kitchen and industrial scales are full bridges.

Quarter, half and full bridge



- quarter: $V_d \approx \frac{V_B}{4} x$ half: $V_d = \frac{V_B}{2} x$ full: $V_d = V_B\, x$ $(x = k\,\varepsilon)$
- half and full bridge are exactly linear: the leg current stays constant
- temperature changes all gauges equally: compensated

▶ Falstad: `examples/bridges/bridge_types.txt`

In the half bridge the left leg contains $R(1-x)$ and $R(1+x)$; the sum is $2R$, so the leg current does not change, and the tap voltage is $V_B(1+x)/2$, exactly linear. In the full bridge the right leg moves in the opposite direction and doubles the output once more. A temperature change alters all four gauges by the same factor and leaves the ratios, and thus the output, unchanged. In a quarter bridge the same effect is obtained with an unstrained *dummy gauge* on the same material as R_4.

Exercise 5.2: Strain gauge bridge

Strain gauges with $R = 350\,\Omega$ and $k = 2$ are glued to a beam. The strain is $\varepsilon = 1000\,\mu\text{m/m}$, the bridge supply $V_B = 5\,\text{V}$.

a) Calculate ΔR.

b) Calculate the output of a quarter, a half and a full bridge.

c) Which gain is needed to get $2\,\text{V}$ at this strain from the full bridge?

5.4 Resistive Temperature Sensors

The resistance of metals rises with temperature. Platinum is used for precision sensors because it is chemically stable and its characteristic is reproducible and standardised (IEC 60751). A Pt100 has $100\,\Omega$ at $0\,^\circ\text{C}$, a Pt1000 $1000\,\Omega$. Between $0\,^\circ\text{C}$ and $100\,^\circ\text{C}$ the mean temperature coefficient is $\alpha = 3.85 \times 10^{-3}/\text{K}$, so a Pt100 changes by $0.385\,\Omega/\text{K}$. Over wider ranges the Callendar–Van Dusen equation describes the slight curvature. Thermistors (NTC: negative temperature coefficient) are semiconductor sensors: cheap, ten times more sensitive, but strongly nonlinear and less accurate.

Platinum sensors Pt100 and Pt1000

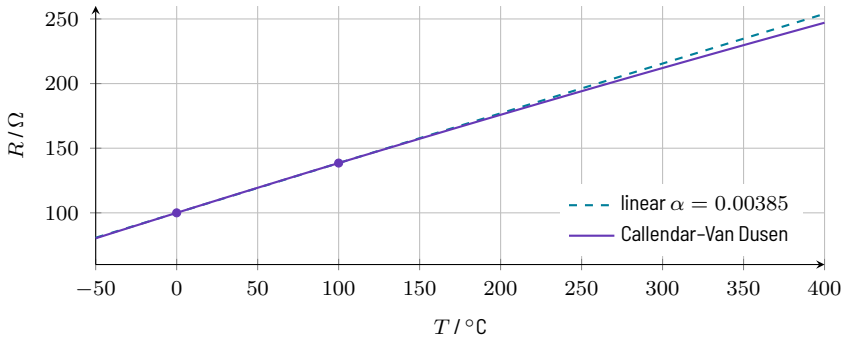
- linear model ($0\,°\text{C} \ldots 100\,°\text{C}$):

Pt100 / Pt1000

$$R(T) = R_0\,(1 + \alpha T), \quad R_0 = 100\,\Omega \,/\, 1000\,\Omega, \quad \alpha = 3.85 \times 10^{-3}/\text{K}$$

- Pt100: $100\,\Omega$ at $0\,°\text{C}$, $138.5\,\Omega$ at $100\,°\text{C}$, $0.385\,\Omega/\text{K}$
- Callendar–Van Dusen ($T \geq 0\,°\text{C}$): $R(T) = R_0(1 + AT + BT^2)$
  - $A = 3.9083 \times 10^{-3}/\text{K}$, $B = -5.775 \times 10^{-7}/\text{K}^2$
- tolerance class B: $\pm(0.3\,\text{K} + 0.005\,|T|)$, class A: $\pm(0.15\,\text{K} + 0.002\,|T|)$

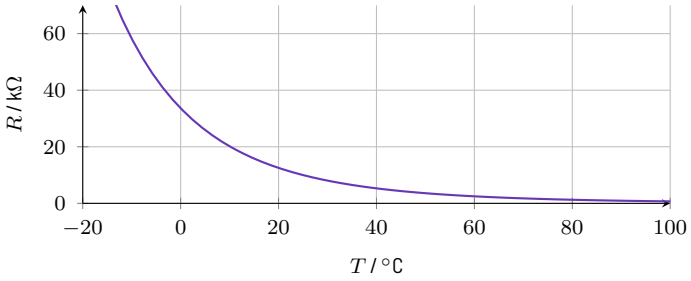
Pt100 characteristic



- $0\,°\text{C}$ … $100\,°\text{C}$: linear model and standard agree at both ends, $0.4\,\text{K}$ apart at $50\,°\text{C}$
- $400\,°\text{C}$: $254\,\Omega$ (linear) vs. $247\,\Omega$: $18\,\text{K}$ error

For $T < 0\,°\text{C}$ the Callendar–Van Dusen equation has an additional term $C\,(T - 100\,°\text{C})\,T^3$ with $C = -4.183 \times 10^{-12}/\text{K}^4$. At $50\,°\text{C}$ the standard gives $119.40\,\Omega$, the linear model $119.25\,\Omega$; the difference of $0.15\,\Omega$ corresponds to $0.4\,\text{K}$. For the capstone with its resolution of $12.5\,\text{K}$ per LSB the linear model is more than good enough.

NTC thermistors



- $R(T) = R_{25}\,\exp\left(B\left(\frac{1}{T} - \frac{1}{T_{25}}\right)\right)$, e.g. $R_{25} = 10\,\text{k}\Omega$, $B = 3950\,\text{K}$ (T in kelvin)
- sensitivity at $25\,°\text{C}$: $-B/T^2 = -4.4\,\%/\text{K}$ (Pt: $+0.39\,\%/\text{K}$)
- strongly nonlinear: linearise with a series resistor, or a table in software

Self-heating: why the sensor current is limited

- the measuring current heats the sensor: $P = I^2 R$

- self-heating coefficient E: e.g. 0.1 K/mW (thin film, in water) … 1 K/mW (in still air)

  Self-heating error
  $\Delta T = E \cdot I^2 R$

- Pt100, 1 mA: $P = 0.1$ mW, $\Delta T = 0.01$ K … 0.1 K
- Pt100, 10 mA: $P = 10$ mW, $\Delta T = 1$ K … 10 K
- rule: Pt100 ≤ 1 mA, Pt1000 ≤ 0.3 mA (capstone: $I \leq 1$ mA)

The sensor measures its own temperature, not that of the medium. As long as the self-heating is constant it could be calibrated out, but it depends on how well the sensor is coupled to the medium: a sensor in flowing water is cooled much better than the same sensor in still air. A smaller current reduces the error with the square, but also the signal linearly. For a Pt1000 the same power is reached at a $\sqrt{10}$ times smaller current.

Exercise 5.3: Pt100 values

a) Calculate the resistance of a Pt100 at $25\,^\circ$C and $37\,^\circ$C with the linear model.

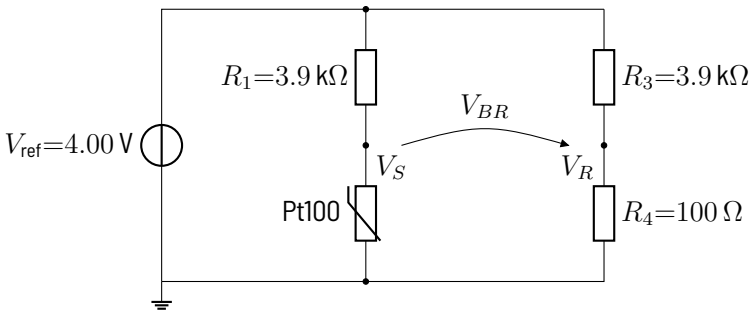
b) Calculate $R(100\,^\circ\text{C})$ and $R(200\,^\circ\text{C})$ with the Callendar–Van Dusen equation. Which temperature does the linear model assign to these resistances?

c) A Pt100 is read with 5 mA in still air ($E = 0.5$ K/mW). How large is the self-heating error at $0\,^\circ$C?

5.5 Designing the Sensor Bridge

We now design the bridge of the capstone task. The requirements: a Pt100 for $0\,^\circ$C … $100\,^\circ$C, a sensor current of at most 1 mA, zero output at $0\,^\circ$C, and an amplifier that produces exactly 4.00 V at $100\,^\circ$C. The bridge is excited from the 4.00 V reference that also feeds the ladder of the flash converter, so that the measurement is ratiometric (chapter 4). The same design is used in the capstone chapter.

The capstone bridge



- $R_4 = R_0 = 100\,\Omega$: balanced at $0\,°\text{C}$
- $I = V_{\text{ref}}/(R_1 + R) \le 1\,\text{mA} \Rightarrow R_1 \ge 3.9\,\text{k}\Omega$; $R_3 = R_1$ ($m = 39$)
- $V_{BR}(T) = V_{\text{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right)$

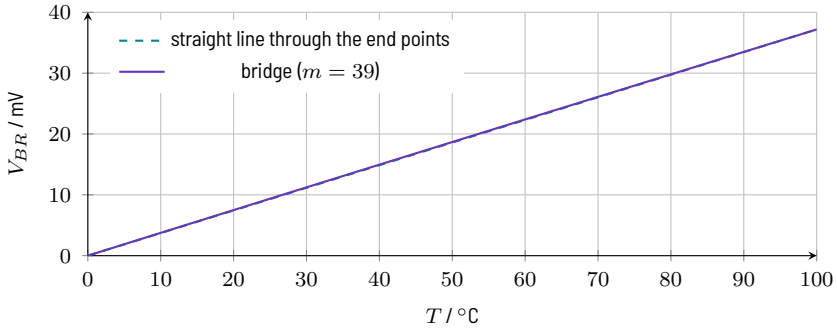
Numbers of the capstone bridge

	$0\,°\text{C}$	$50\,°\text{C}$	$100\,°\text{C}$
sensor R	$100\,\Omega$	$119.25\,\Omega$	$138.5\,\Omega$
sensor current	$1.000\,\text{mA}$	$0.995\,\text{mA}$	$0.990\,\text{mA}$
V_S	$100.0\,\text{mV}$	$118.7\,\text{mV}$	$137.2\,\text{mV}$
V_R	$100.0\,\text{mV}$	$100.0\,\text{mV}$	$100.0\,\text{mV}$
V_{BR}	$0\,\text{mV}$	$18.68\,\text{mV}$	$37.18\,\text{mV}$

- sensitivity $\approx 0.37\,\text{mV/K}$; nonlinearity at $50\,°\text{C}$: $+0.09\,\text{mV}$ ($0.25\,\%$ of full scale)
- required gain: $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- self-heating: $P = 0.1\,\text{mW}$

▶ Falstad: `examples/bridges/pt100_bridge.txt`

Bridge output over temperature



- with $m = 39$ the curve is almost straight: maximum deviation $0.09\,\text{mV} \approx 0.24\,\text{K}$
- with $m = 1$ (all arms $100\,\Omega$) the deviation would be $4.4\,\text{K}$

Design card

Design card: Sensor bridge

- excitation V_B: the ADC reference (ratiometric), here $4.00\,\text{V}$
- reference arm: $R_4 = R_{\text{sensor}}(T_{\text{min}}) \Rightarrow V_d = 0$ at T_{min}
- current limit: $R_1 \geq V_B / I_{\text{max}} - R_{\text{sensor}}$, round up to E-series; $R_3 = R_1$
- full-scale output $V_d(T_{\text{max}})$ with the exact formula, gain $G = V_{\text{FS}} / V_d(T_{\text{max}})$
- check nonlinearity ($\approx x/(m+1)$), self-heating ($I^2 R \cdot E$), common-mode voltage ($\approx V_B/(m+1)$)

Exercise 5.4: Design of the capstone bridge

A Pt100 ($\alpha = 3.85 \times 10^{-3}/\text{K}$) shall measure $0\,^\circ\text{C} \ldots 100\,^\circ\text{C}$. The bridge is fed from $V_{\text{ref}} = 4.00\,\text{V}$, the sensor current must not exceed $1\,\text{mA}$.

a) Choose R_1, R_3 and R_4.

b) Calculate the bridge output at $0\,^\circ\text{C}$, $50\,^\circ\text{C}$ and $100\,^\circ\text{C}$ and the sensor currents.

c) How large is the deviation from a straight line at $50\,^\circ\text{C}$, in millivolt and in kelvin?

d) Which total amplifier gain gives $4.00\,\text{V}$ at $100\,^\circ\text{C}$?

Exercise 5.5: Alternative: Pt1000 at 5 V

A Pt1000 is used in a bridge with $V_B = 5\,\text{V}$, $R_1 = R_3 = 4.7\,\text{k}\Omega$, $R_4 = 1\,\text{k}\Omega$.

a) Calculate the sensor current and the self-heating power at $0\,^\circ\text{C}$ and $100\,^\circ\text{C}$.

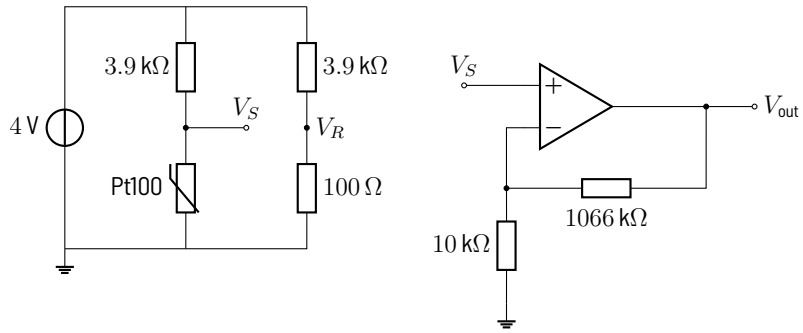
b) Calculate the bridge output at $100\,^\circ\text{C}$ and the required gain for $4.00\,\text{V}$.

c) How large is the deviation from the straight line at $50\,^\circ\text{C}$? Compare with the Pt100 design.

5.6 Reading the Bridge: Common-Mode Voltage

The bridge output is the difference of two node voltages, each of which is far from zero. In the capstone bridge both nodes sit at about $100\,\text{mV}$; with $m = 1$ they would be at half the supply. The mean of the two node voltages is the *common-mode voltage* $V_{\text{CM}} = (V_S + V_R)/2$, the difference is the *differential voltage* $V_d = V_S - V_R$. The amplifier must amplify V_d by about 100 and ignore V_{CM}. An ordinary amplifier referenced to ground cannot do this.

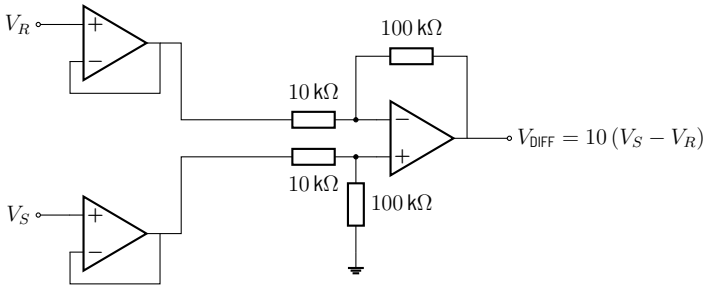
A single-ended amplifier fails



- gain 107.6 applied to $V_S = 100$ mV: $V_{out} = 10.75$ V at 0 °C (wanted: 0 V)
- at 100 °C: 14.76 V; the useful signal is a small change on a large offset
- any drift of V_{ref} appears amplified at the output

▶ Falstad: `examples/bridges/bridge_single_ended.txt`

Difference and instrumentation amplifier



- difference amplifier: amplifies $V_S - V_R$, rejects V_{CM} (if the resistors match)
- buffers: no current from the bridge, the bridge is not loaded
- second stage $G_2 = 10.76$: 0 V … 4.00 V (details: chapter 6)

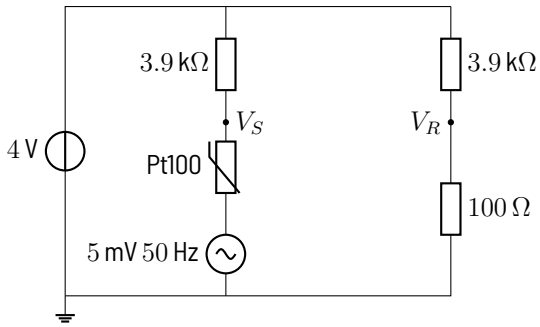
▶ Falstad: `examples/bridges/bridge_ina.txt`

Without the buffers the difference amplifier would load the bridge with its input resistors. Here the bridge has a source resistance of only $3.9\,\text{k}\Omega \parallel 100\,\Omega = 97.5\,\Omega$, so $10\,\text{k}\Omega$ inputs would cause about $1\,\%$ gain error, and worse, a mismatch between the two inputs that converts common-mode voltage into an output error. The two buffers and the difference amplifier together form the classic three-op-amp instrumentation amplifier (INA), treated in chapter 6. In Falstad the complete front end delivers 0 V, 2.01 V and 4.00 V at 0 °C, 50 °C and 100 °C.

5.7 Interference and Lead Resistance

The sensor is often far away from the electronics, connected by a cable of several metres. The cable brings two problems. First, the mains wiring and the fluorescent lamps in the lab create 50 Hz electric and magnetic fields. The loop formed by the two sensor wires acts as the secondary winding of a transformer, and the induced voltage appears in series with the sensor: a differential signal that the instrumentation amplifier amplifies just like the measurement signal. Interference that couples equally into both bridge nodes is common-mode and rejected. In the capstone the overhead lamps are modelled by a 5 mV, 50 Hz source in series with the sensor. Second, the resistance of the cable adds to the sensor resistance. For a Pt100 one ohm corresponds to 2.6 K.

50 Hz in series with the sensor

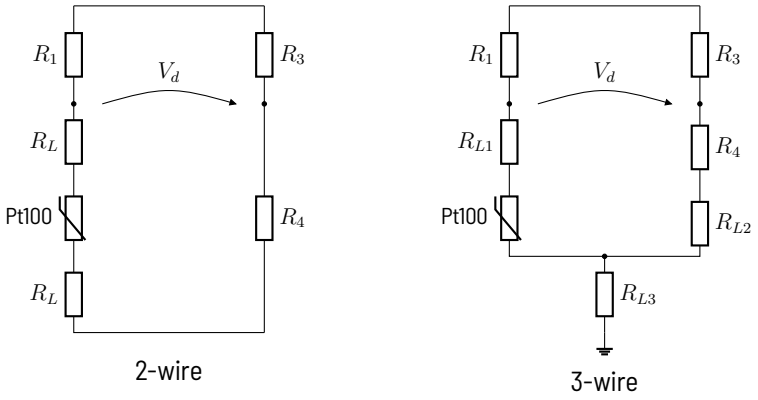


- at V_S: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} = 5\,\text{mV} \cdot \frac{3900}{4000} = 4.88\,\text{mV}$ amplitude
- after $G = 107.6$: $0.52\,\text{V}$ amplitude: about one LSB ($0.5\,\text{V}$)
- remedies: twisted pair, shield, short leads, low-pass filter before the ADC (chapters 6, 7)

▶ Falstad: `examples/bridges/bridge_50hz.txt`

Twisting the two sensor wires reduces the loop area and makes the induced voltages of neighbouring twists cancel. A shield connected to ground at the amplifier side diverts capacitively coupled currents. What remains must be removed by filtering: the temperature changes slowly, so everything above a few hertz can be cut off. Chapter 7 designs the anti-aliasing filter and chapter 8 shows what happens if the interference reaches the sampler unfiltered.

Lead resistance: 2-wire and 3-wire connection



- 2-wire: $2R_L$ in series with the sensor; $R_L = 0.5\,\Omega$: $1\,\Omega \mathrel{\hat{=}} 2.6\,\text{K}$
- 3-wire: R_{L1} in the sensor arm, R_{L2} in the neighbouring arm R_4: they cancel at balance; R_{L3} only carries the supply current
- 4-wire: current source through two wires, voltage sensed by the other two (precision instruments)

▶ Falstad: `examples/bridges/bridge_3wire.txt`

Exercise 5.6: Interference and lead resistance

The capstone bridge ($V_\text{ref} = 4\,\text{V}$, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100) is followed by an amplifier with gain 107.6.

a) A $5\,\text{mV}$, $50\,\text{Hz}$ interference voltage is induced in series with the sensor. Calculate its amplitude at V_S and at the amplifier output. Why does the instrumentation amplifier not reject it?

b) The sensor is connected with two leads of $0.5\,\Omega$ each. Which bridge output and temperature error result at $0\,°$C?

c) Repeat b) for the 3-wire connection, at $0\,°$C and $100\,°$C.

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